

E-Commerce Sales Analysis

Dashboard Sections

The Dashboard consist of main six sections, each addressing a specific analytical goal

1. Sales Overview

This section will provide an overall view of a e-commerce sales performance, including total sales, orders, quantity sold, average order value, sales growth, and sales trends over time.

2. Product & Category Analysis

This section will provide an overall view of a e-commerce sales performance, including total sales, orders, quantity sold, average order value, sales growth, and sales trends over time.

3. Customer Analysis

This section will provide insights into customer behavior, customer segments, spending patterns, order activity, and the contribution of different customer groups to overall sales.

4. Geographical Analysis

This section will analyze sales and order performance across different states and cities to identify high and low performing geographical regions.

5. Order & Delivery Analysis

This section will monitor order statues, cancellation and return rates, payment methods, and delivery performance, including average delivery days.

Data Requirements

Sales Overview

- Order ID
- Order Date
- Customer ID
- Product ID
- Quantity
- Total Amount / Sales
- Unit Price
- Coupon Discount
- Order Status
- Payment Mode
- Category
- State

Product & Category Analysis

- Product ID
- Product Name
- Category
- Brand
- Original Price
- Selling Price
- Discount %
- Stock Quantity
- Average Rating
- Total Reviews
- Quantity Sold
- Sales / Total Amount

Customer Analysis

- Customer ID
- Customer Name
- Gender
- Age
- Age Group
- Customer Segment
- City
- State
- Total Orders
- Total Spent
- Order Date
- Sales / Total Amount

Geographical Analysis

- State
- City
- Customer ID
- Order ID
- Sales / Total Amount
- Quantity
- Order Status

Order & Delivery Analysis

- Order ID
- Order Date
- Delivery Date
- Order Status
- Payment Mode
- Shipping Cost
- Quantity
- Total Amount
- State
- City

Filters / Slicers

- Date
- Category
- Brand
- State
- Customer Segment
- Order Status
- Payment Mode

Visuals / Charts

Sales Overview

- **KPI Cards** (Total Sales, Total Orders, Total Quantity Sold, Total Customers, Average Order Value, Sales Growth)
- **Line Chart** (Monthly Sales Trend)
- **Column Chart** (Sales by Category)
- **Donut Chart** (Order Status)
- **Pie Chart** (Sales by Order State)
- **Bar Chart** (Top 10 States by Sales)

Product & Category Analysis

- **KPI Cards** (Total Products, Total Sales, Total Quantity Sold, Average Product Rating)
- **Column Chart** (Sales by Category, Quantity Sold by Category)
- **Bar Chart** (Top 10 Products by Sale, Sales by Brand, Product Rating Analysis)
- **Scatter Chart** (Sales by Discount %)

Customer Analysis

- **KPI Cards** (Total Customers, Average Customer Spending, Average Order per Customer, Average Order per Customer)
- **Column Chart** (Sales by Customer Segment, Customers by Age-group, Sales by Age-group)
- **Bar Chart** (Top 10 Customers by Spending)
- **Scatter Chart** (Customer Orders vs Spending)
- **Donut Chart** (Sales by Gender)

Geographical Analysis

- **KPI Cards** (Total Sales, Total Orders, Total Customers, Average Sales per Customer)
- **Column Chart** (Orders by State, Quantity Sold by State)
- **Bar Chart** (Sales by State, Top 10 Cities by Sale)

Order & Delivery Analysis

- **KPI Cards** (Total Orders, Cancellation Rate, Return Rate, Average Delivery Days, Shipping Cost)
- **Column Chart** (Orders by Payment Mode)
- **Bar Chart** (Average Delivery Days)
- **Line Chart** (Orders by Month, Cancelled Orders by Month)
- **Donut Chart** (Orders by Status)

Interactivity

The dashboard will provide interactive features that allow users to explore and analyze e-commerce data dynamically.

1. Interactive Slicers – Users can filter the dashboard by Date, Category, Brand, State, Customer Segment, Order Status, Payment Mode, Gender, Age Group, and City.
2. Cross-Filtering – Selecting a data point in one visual will automatically filter related visuals on the same page.
3. Cross-Highlighting – Users can select a category, product, customer segment, or region to highlight its contribution in other charts.
4. Drill-Down – Users can move from higher-level information to detailed information, such as Year → Quarter → Month or State → City.
5. Drill-Through – Users can right-click a product, customer, or region and navigate to a dedicated detailed analysis page.
6. Tool tips – Hovering over charts or data points will display additional information such as sales, quantity, orders, and percentages.
7. Page Navigation – Users can navigate between the five dashboard sections: Sales Overview
 - Sales Overview
 - Product & Category Analysis
 - Customer Analysis
 - Geographical Analysis
 - Order & Delivery Analysis
8. Dynamic KPI Updates – KPI cards and charts will automatically update based on the filters and selections made by the user.
9. Reset Filters – A reset/clear-filter option can be provided to return the dashboard to its default view.

Measures

1. Total Sales

Calculate the sum of all the sales recorded

DAX: Total Sales = SUM(Sales[Total_Amount])

2. Total Orders

Calculate the total number of unique orders placed

DAX: Total Orders = DISTINCTCOUNT(Sales[Order_ID])

3. Total Quantity Sold

Calculates the total number of products sold.

DAX: SUM(Sales[Quantity])

4. Average Order Value (AOV)

Calculates the average amount spent per order.

DAX: DIVIDE([Total Sales], [Total Orders])

5. Total Discount

Calculates the total discount provided to customers through coupons.

DAX: SUM(Sales[Coupon_Discount])

6. Total Shipping Cost

Calculates the total shipping cost associated with orders.

DAX: SUM(Sales[Shipping_Cost])

7. Total Customers

Calculates the total number of unique customers who placed orders.

DAX: DISTINCTCOUNT(Sales[Customer_ID])

8. Average Customer Spending

Calculates the average sales amount generated per customer.

DAX: `DIVIDE([Total Sales], [Total Customers])`

9. Average Orders per Customer

Calculates the average number of orders placed by each customer.

DAX: `DIVIDE([Total Orders], [Total Customers])`

10. Cancelled Orders

Calculates the total number of cancelled orders.

DAX: `CALCULATE([Total Orders], Sales[Order_Status] = "Cancelled")`

11. Cancellation Rate

Calculates the percentage of orders that were cancelled.

DAX: `DIVIDE([Cancelled Orders], [Total Orders])`

12. Returned Orders

Calculates the total number of returned orders, if returned is represented in the order status.

DAX: `CALCULATE([Total Orders], Sales[Order_Status] = "Returned")`

13. Return Rate

Calculates the percentage of orders that were returned.

DAX: `DIVIDE([Returned Orders], [Total Orders])`

14. Average Delivery Days

Calculates the average number of days taken to deliver orders.

DAX: `AVERAGE(Sales[Delivery Days])`

15. Maximum Delivery Days

Identifies the longest delivery time recorded.

DAX: `MAX(Sales[Delivery Days])`

16. Minimum Delivery Days

Identifies the shortest delivery time recorded.

DAX: `MIN(Sales[Delivery Days])`

17. Previous Month Sales

Calculates sales generated during the previous month.

DAX: `CALCULATE([Total Sales], DATEADD(Dim_Date[Date], -1, MONTH))`

18. Sales Growth %

Measures the percentage change in sales compared with the previous month.

DAX: `DIVIDE([Total Sales] - [Previous Month Sales], [Previous Month Sales])`

19. Previous Year Sales

Calculates sales generated during the corresponding previous-year period.

DAX: `CALCULATE([Total Sales], SAMEPERIODLASTYEAR(Dim_Date[Date]))`

20. YoY Growth %

Measures the percentage change in sales compared with the same period in the previous year.

DAX: `DIVIDE([Total Sales] - [Previous Year Sales], [Previous Year Sales])`

21. Total Products

Calculates the total number of unique products in the product dataset.

DAX: `DISTINCTCOUNT(Products[Product_ID])`

22. Average Product Rating

Calculates the average rating of the products.

DAX: `AVERAGE(Products[Avg_Rating])`

Export Needs

- **Export Options:** PDF, Excel, CSV, Power BI Service shareable link
- **Distribution:** Management and internal analytics team